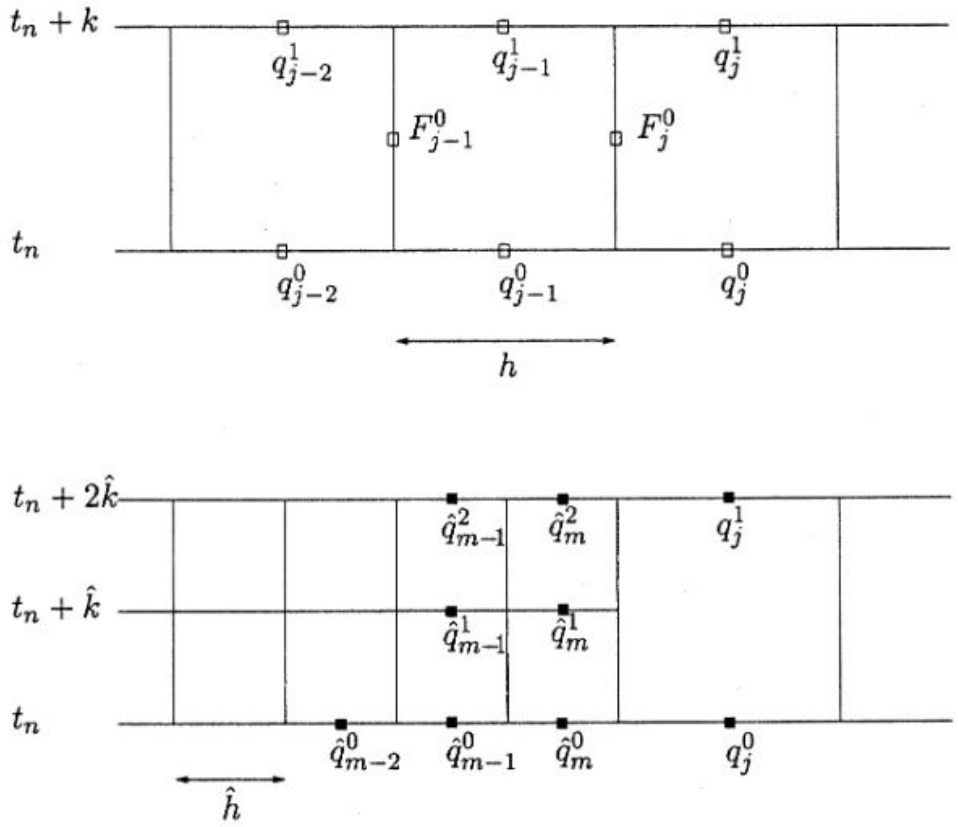
Adaptive Mesh Refinement

Principle of mesh refinement

Refinement

- a tagged cell of the original grid at level *L* is divided in R^L cells of the same size

- The time step of the original grid divided by R^L *is used to update the variables on the refined cells at level L*



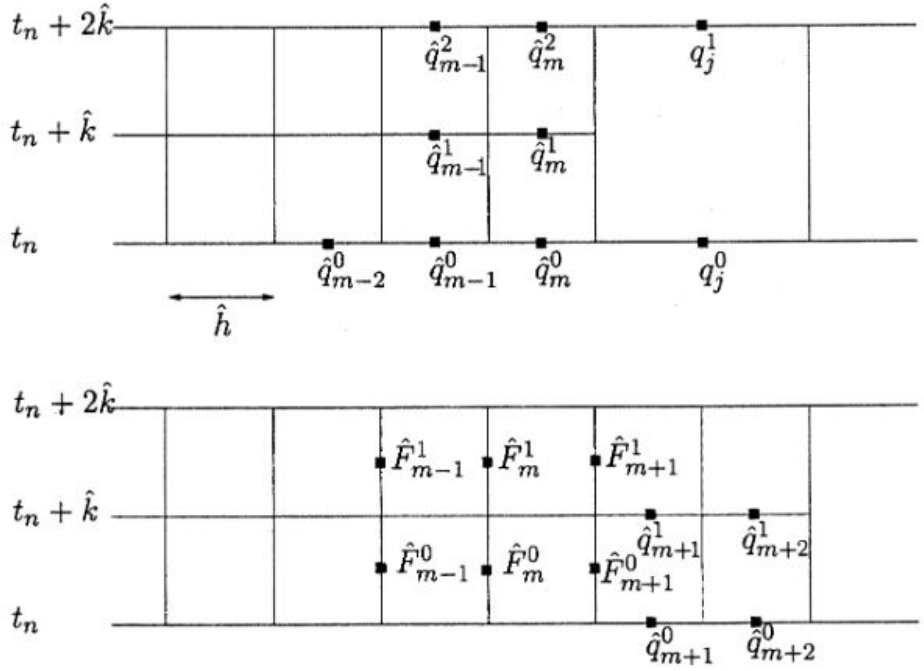
FIGURE, Marsha J. Berger and Randall J. Leveque, Adaptive Mesh Refinement Using Wave-Propagation Algorithms For Hyperbolic Systems. Siam J Number Anal's, v 35, No 36, p 2298-2316, 1998.

Principle of mesh refinement

Boundary Conditions

- Boundary Conditions are imposed using ghost cells.

- Fine grid may have interior boundaries with coarser grid. In this case the values of the ghost cells are interpolated from the coarse grid.



FIGURE, Marsha J. Berger and Randall J. Leveque, Adaptive Mesh Refinement Using Wave-Propagation Algorithms For Hyperbolic Systems. Siam J Number Anal's, v 35, No 36, p 2298-2316, 1998.

Refinement Criterion

Error estimator

Usually the refinement criteria is based on an error estimator. There are two types of error estimator:

A priori

- Evaluate where the most errors is made before the time step.
- Depends on the exact solution (usually the gradients of the solution)

A posteriori

- Evaluate where the most errors is made after the time step.
- Depends on the computing solutions.

Refinement Criterion

Example of error estimator

A priori estimator:

In *Adaptive mesh refinement for two-phase slug flows with an a priori indicator* from M. Gourma, N. Jia and C; Thompson, a half grid of level *L* is refined if

$$\overline{\nabla U_{l,\pm}}^2 \geq \eta_l \quad \text{where} \quad \begin{cases} \overline{\nabla U_{l,-}}^2 = 2 \sum_{i=2}^{nl/2} (U_i - U_{i-1})^2 / nl \\ \overline{\nabla U_{l,+}}^2 = 2 \sum_{i=nl/2+1}^{nl} (U_i - U_{i-1})^2 / nl \end{cases} \qquad \eta_l = \left(\sum_{k=1}^{l} \overline{\nabla U_{k,-}}^2 + \overline{\nabla U_{k,+}}^2 \right)^{\frac{1}{2}} \Big/ \sqrt{2Nl}$$

Refinement Criterion

Example of error estimator

A posteriori estimator:

The error is estimated using the truncation error between to different level.

Richardson extrapolation is comparing the computed solutions using two time steps. One computed using the original grid and the other using a grid twice as coarser.